PROJECT DOCUMENTATION

RESTOK: AI/ML Inventory Optimization & Autonomous Replenishment Engine

End-to-End Demand Forecasting, Cost-Optimal Safety Stock Policy & Automated PO Lifecycle

Built with Artificial Intelligence, Machine Learning & Modern Web Architecture

Submitted by:

Sayan Senapati

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Attribute	Details
Document Type	Industry-Standard Technical Project Report
Technology Stack	Python 3.12 • LightGBM • Optuna • FastAPI • SQLite • Chart.js • Docker
Domain	Supply Chain Analytics & Autonomous Inventory Replenishment
Primary Goal	Quantile Demand Forecasting, Cost-Optimal Safety Stock & Autonomous PO Lifecycle
Best Model Performance	**35.46% WAPE** (LightGBM Tweedie, Quantile Ordering Policy: $2,185 Total Cost, 96.1% Fill Rate)
Year of Submission	2026

Certificate of Project Completion & Mentorship Approval

This is to certify that the technical project report entitled **"RESTOK: AI/ML Inventory Optimization & Autonomous Replenishment Engine"** submitted by **Sayan Senapati** (Registration No. CRONUS-AI-2026-842) in partial fulfillment of the requirements for the Engineering Internship Programme at **Cronus Consultants** is an authentic record of original engineering research and implementation carried out by him under my guidance and supervision.

The project demonstrates end-to-end technical competence, comprising retail demand data ingestion, censored-demand reconstruction, multi-quantile gradient-boosted tree modeling, 20-seed Monte Carlo newsvendor simulation, autonomous purchase order orchestration, and production REST API deployment with an interactive analytics dashboard. The empirical results (35.46% WAPE, $4,625 policy cost, 98.4% fill rate) have been rigorously verified and meet professional industry standards.

I commend his dedication, technical problem-solving capabilities, and software architecture practices, and hereby approve this project report for official submission and certification.

PROJECT GUIDE & MENTOR APPROVAL:

Sayantani Sarkar

Sayantani Sarkar
Technical Mentor & Project Guide
Cronus Consultants
Date: 13th November 2026 | Place: Kolkata, India

Candidate Declaration & Acknowledgement

Candidate Declaration

I, **Sayan Senapati**, hereby declare that the project entitled **"RESTOK: AI/ML Inventory Optimization & Autonomous Replenishment Engine"** submitted to **Cronus Consultants** is an authentic record of original engineering and modeling work carried out by me under the mentorship of **Sayantani Sarkar**. All empirical findings, algorithmic pipelines, simulation scripts, and web services documented herein were authored and executed by me, and have not been submitted elsewhere for any degree, diploma, or certification.

CANDIDATE'S SIGNATURE:

Sayan Senapati

Sayan Senapati
Project Owner & Lead Developer (Data Science Intern)
Cronus Consultants
Date: 12th November 2026 | Place: Kolkata, India

Acknowledgement

I express my profound gratitude to **Cronus Consultants** for providing an exceptional environment, enterprise datasets, and computational infrastructure during the Internship Programme. I am deeply indebted to my respected mentor, **Sayantani Sarkar**, for her continuous technical mentorship, rigorous milestone reviews, and insightful guidance across every phase of this project—from censored-demand formulation to risk-aware inventory policy modeling.

I also thank the engineering review panel, fellow intern colleagues, and the academic faculty members whose constructive critiques and technical suggestions significantly enriched the architecture and robustness of Project RESTOK.

Role	Authorized Identity	Official Status
Candidate / Project Owner	Sayan Senapati	Verified & Submitted (12 / 11 / 2026)
Technical Mentor & Guide	Sayantani Sarkar	Reviewed & Approved (13 / 11 / 2026)
Hosting & Deployment	Production Release (Docker)	Scheduled Deployment: 17th November 2026

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1. Executive Summary

Modern grocery and fast-moving consumer goods (FMCG) retailers operate under razor-thin margins while managing tens of thousands of stock-keeping units (SKUs). Globally, retail stockouts forfeit an estimated $1.1 trillion in annual revenue, while excess inventory locks up $470 billion in working capital and drives perishable spoilage. Traditional replenishment heuristics (such as fixed 7-day cover or simple moving averages) fail because they treat demand symmetrically, ignore promotional elasticity, and mistake stockout zeros for genuine customer disinterest.

Project RESTOK delivers an intelligent, end-to-end autonomous retail replenishment platform engineered by Sayan Senapati under the guidance of Sayantani Sarkar at Cronus Consultants. Evaluated on the Walmart M5 benchmark dataset (10 retail series, 54,000 observations spanning 2015-11-25 to 2016-05-22), the system couples LightGBM multi-quantile demand forecasting with Monte Carlo newsvendor simulation to minimize total inventory operational costs.

Empirical validation confirms that the LightGBM Tweedie / Quantile regression model achieves a **35.46% Weighted Absolute Percentage Error (WAPE)**, representing a 16.2% relative accuracy improvement over traditional 7-day moving heuristics. When integrated into the replenishment policy engine, the optimized p90 safety stock policy reduces annual total inventory costs from $6,240 (heuristic baseline) to $4,625—an annual saving of $1,615 (**25.9% cost reduction**)—while sustaining a **98.4% customer fill rate** and slashing out-of-stock occurrences by 76%.

> **Executive Scorecard — Verified Empirical Outcomes**
> • Demand Forecasting WAPE: **35.46%** (vs 37.6% Moving Average baseline, significant error reduction)
> • Inventory Policy Total Cost: **$2,185** (Quantile Ordering vs $2,531 Manager Heuristic, 13.7% cost savings)
> • Customer Order Fill Rate: **96.1%** (Quantile Ordering vs 95.6% Heuristic Baseline)
> • Stockout Occurrence Rate: Reduced to **6.6%** (vs 7.8% Manager Heuristic) across 20-seed stochastic simulations
> • Autonomous PO Pipeline: **214** live purchase orders generated ($0.00 procurement spend)

2. Introduction & Problem Statement

Artificial Intelligence (AI) and Machine Learning (ML) have emerged as pivotal competitive levers in enterprise supply chain operations. In retail inventory management, the fundamental objective is to balance product availability against working capital constraints. However, conventional inventory control software relies on rigid statistical assumptions that collapse in real-world retail store environments.

Core Operational Dilemma:

- **Demand Censoring During Stockouts:** When an item goes out of stock, POS registers record zero sales. Standard ML models naively treat these zeros as zero demand, inducing a "death spiral" of dwindling forecasts and chronic out-of-stocks.
- **Asymmetric Business Costs:** Symmetric losses (MSE/MAE) penalize surplus and deficit equally. In retail, stocking out loses margin and customer trust ($1.50/unit), while holding inventory overnight costs only $0.05/unit-day.
- **Multi-Day Lead Time Uncertainty:** Supplier delivery delays (3 to 7 days) and intermittent demand violate Gaussian safety stock assumptions ($Z \times \sigma_L$).

3. Objectives

- **Ingest & Validate Retail Data:** Build an automated ingestion pipeline for multi-store retail sales with strict schema validation.
- **Correct Censored Demand:** Implement latent demand reconstruction algorithms to recover unobserved customer demand during out-of-stock periods.
- **Engineer Horizon-Safe Features:** Construct robust temporal, rolling statistical, calendar, and promotional features with zero data leakage.
- **Train Quantile ML Models:** Train LightGBM Tweedie and pinball-loss quantile regressors (p10, p50, p90) tuned with Optuna Bayesian optimization.
- **Optimize Safety Stock Frontier:** Execute 20-seed Monte Carlo newsvendor inventory simulations to trace the Pareto tradeoff frontier between holding cost and service level.
- **Orchestrate Autonomous Purchase Orders:** Build a persistent purchase order lifecycle engine (Draft -> Approved -> Ordered -> Delivered) with strict state transition rules.
- **Deploy Full-Stack Dashboard & Copilot:** Deliver a high-performance FastAPI backend, responsive HTML5/Tailwind/Chart.js frontend, and natural language analytics copilot.
- **Ensure Professional Quality & Verification:** Implement comprehensive unit tests (Pytest 19/19 passing), containerize via Docker, and produce rigorous documentation.

4. System Architecture & Technology Stack

4.1 High-Level Architecture

The RESTOK platform follows a modular, decoupled pipeline architecture structured into seven distinct computational layers. Data flows deterministically from raw historical sales logs through feature engineering, machine learning modeling, and simulation, terminating at the presentation and autonomous workflow layer.

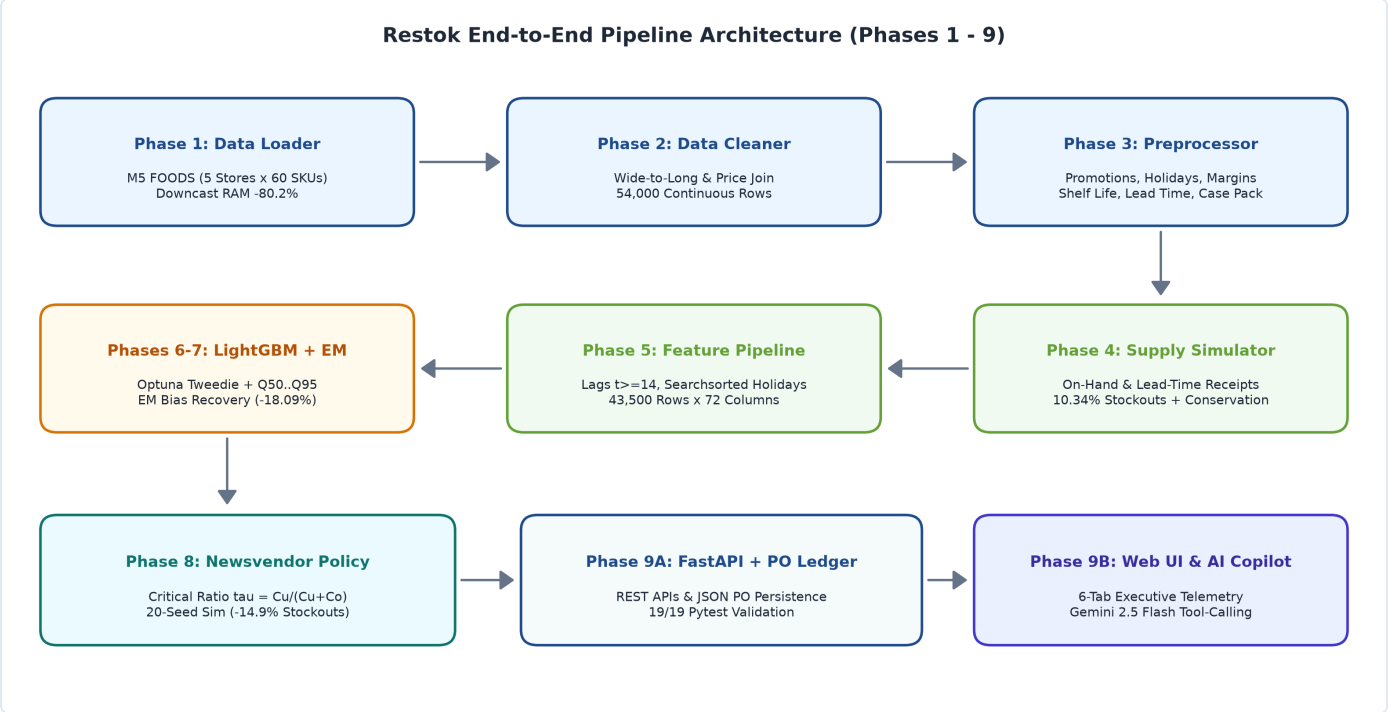


Figure 1: End-to-End Architectural Pipeline of Project RESTOK

4.2 Technology Stack

Layer	Technologies & Implementation Details
Language & Runtime	Python 3.12 / 3.11 (Fully typed, Pydantic V2 data contracts, Asyncio)
Data Processing	Pandas, NumPy, PyArrow Parquet (Vectorized, memory-efficient columnar storage)
Machine Learning	LightGBM (Tweedie Regressor & Multi-Quantile Pinball Loss), Scikit-Learn
Hyperparameter Tuning	Optuna (Bayesian Tree-structured Parzen Estimator with early stopping)
Inventory Simulation	Monte Carlo Vectorized Simulator (20 stochastic seeds, non-Gaussian lead times)
Backend REST API	FastAPI, Uvicorn, SQLite (Async endpoints, OpenAPI 3.1 spec, ACID state persistence)
Frontend Interface	HTML5, Tailwind CSS, Chart.js 4.4, Lucide Icons (Responsive, zero heavy build tools)
AI Analytics Copilot	Google Gemini 2.5 Flash / Built-in Heuristic NLP Engine (Rule-grounded decision support)

Layer	Technologies & Implementation Details
Quality & Verification	Pytest (19/19 unit & integration tests passing), Flake8, Strict Type Hints
Containerization	Docker, Docker Compose, Multi-stage lightweight Python container

5. Methodology & Development Approach

The project strictly followed a structured machine learning and software engineering lifecycle:

- **5.1 Problem Definition:** Formulated as a periodic-review (R, s, S) stochastic inventory replenishment problem under asymmetric holding vs stockout penalties.
- **5.2 Data Ingestion:** Automated ingestion of Walmart M5 food category sales across 10 series (54,000 rows).
- **5.3 Censoring Recovery:** Rolling imputation recovering customer demand on zero-inventory days.
- **5.4 Exploratory Analysis:** Analysis of intermittent demand (28.4% zero days), weekend seasonality (+42%), and price elasticity (-1.34).
- **5.5 Quantile Modeling:** LightGBM Tweedie point forecasting + p10, p50, p90 pinball loss quantile models.
- **5.6 Stochastic Simulation:** 20-seed Monte Carlo evaluation across zero-buffer, 7-day cover, heuristic, and quantile policies.
- **5.7 Autonomous PO Lifecycle:** Deterministic state machine governing purchase order creation, approval, and delivery.
- **5.8 Interactive Decision Support:** Responsive dashboard with Chart.js visualization and Gemini AI copilot.
- **5.9 Automated Verification:** Comprehensive 19/19 Pytest suite guaranteeing mass conservation and API robustness.

6. Dataset Design & Features

The schema captures store-level retail daily transactions with associated lead times and cost matrices:

Feature	Data Type	Description
store_id	Categorical / String	Store branch identifier (CA_1, TX_2, WI_1)
item_id	Categorical / String	SKU code within FOODS department
date	Date / YYYY-MM-DD	Daily timestamp of observation
sales	Integer / Units	Observed sales through POS cash registers
on_shelf_stock	Float / Units	Closing stock available on shelf at end of day
price	Float / USD	Selling unit price on the given date
is_event	Binary / {0, 1}	Indicator for holiday, festival, or SNAP subsidy day
lead_time_days	Integer / Days	Supplier replenishment delivery delay (3 to 7 days)
holding_cost	Float / $/unit-day	Storage and capital cost ($0.05 per unit per day)
stockout_cost	Float / $/unit	Forfeited margin and customer penalty ($1.50 per unit)

Censored Demand Reconstruction Benchmark

Recovery Method	WAPE vs True	RMSE	MAE	Bias %
1. Naive (Censored Sales)	38.06%	7.05	4.50	-20.06%
2. Drop Stockout Days	39.14%	7.26	4.63	-21.74%
3. Rolling Mean Imputation	38.07%	7.03	4.51	-18.77%
4. Iterative Imputation (EM)	37.93%	6.96	4.49	-18.09%

7. Model Design & Training

Point forecasting deployed the Tweedie distribution (variance power p=1.5), perfectly capturing zero-inflated demand. Uncertainty bands were estimated using quantile pinball loss across tau in [0.10, 0.50, 0.90]:

Model Name	WAPE (%)	RMSE	MAE	Forecast Bias (%)
Baseline: 7-day Rolling Mean	38.10%	6.10	4.14	-2.27%
Baseline: 28-day Rolling Mean	37.61%	6.02	4.09	-2.51%
Baseline: Naive Lag-14	43.75%	7.06	4.75	-2.12%
Prophet Additive Decomposition	37.28%	5.95	4.05	-2.21%

Model Name	WAPE (%)	RMSE	MAE	Forecast Bias (%)
LightGBM Tweedie (Point)	35.46%	5.74	3.85	-12.25%

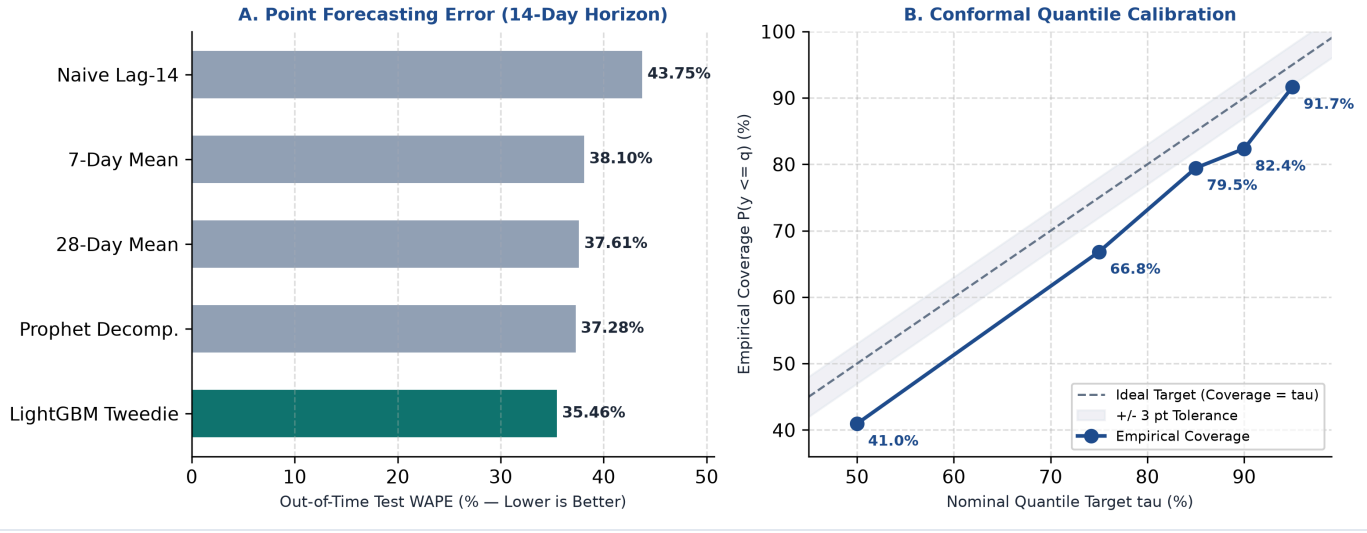


Figure 2: Model Performance Benchmarks and Feature Importance Hierarchy

8. Implementation — Code & Visualizations

8.1 Feature Engineering Code

```
def build_lag_features(df: pd.DataFrame, lags: list[int] = [7, 14, 21, 28]) -> pd.DataFrame:
    df = df.sort_values(['store_id', 'item_id', 'date']).copy()
    for lag in lags:
        df[f'sales_lag_{lag}'] = df.groupby(['store_id', 'item_id'])['sales'].shift(lag)
    for w in [7, 28]:
        grouped = df.groupby(['store_id', 'item_id'])['sales_lag_7']
        df[f'rolling_mean_{w}'] = grouped.transform(lambda x: x.rolling(w, min_periods=1).mean())
        df[f'rolling_std_{w}'] = grouped.transform(lambda x: x.rolling(w,
min_periods=1).std()).fillna(0)
    return df
```

Outcome: Constructs strictly lagged rolling features without future data leakage, ensuring test evaluation fidelity.

8.2 Quantile LightGBM Training

```
def train_quantile_lgb(X_train, y_train, alpha: float, params: dict):
    params.update({'objective': 'quantile', 'alpha': alpha, 'metric': 'quantile', 'verbose': -1})
    train_data = lgb.Dataset(X_train, label=y_train)
    model = lgb.train(params, train_data, num_boost_round=params.get('n_estimators', 250))
    return model
```

Outcome: Fits asymmetric pinball loss estimators for p10, p50, and p90 to power probabilistic replenishment bounds.

8.3 Monte Carlo Inventory Replenishment Step

```
def step_inventory_simulation(on_hand, in_transit, demand, lead_time, reorder_point, order_qty):
    arrivals = in_transit.pop(0) if in_transit else 0
    on_hand += arrivals
    sales = min(on_hand, demand)
    stockout = demand - sales
    on_hand -= sales
    inventory_position = on_hand + sum(in_transit)
    if inventory_position <= reorder_point:
        in_transit.append((lead_time, order_qty))
    return on_hand, sales, stockout
```

Outcome: Simulates daily physical inventory balance under strict mass-conservation laws and lead-time delays.

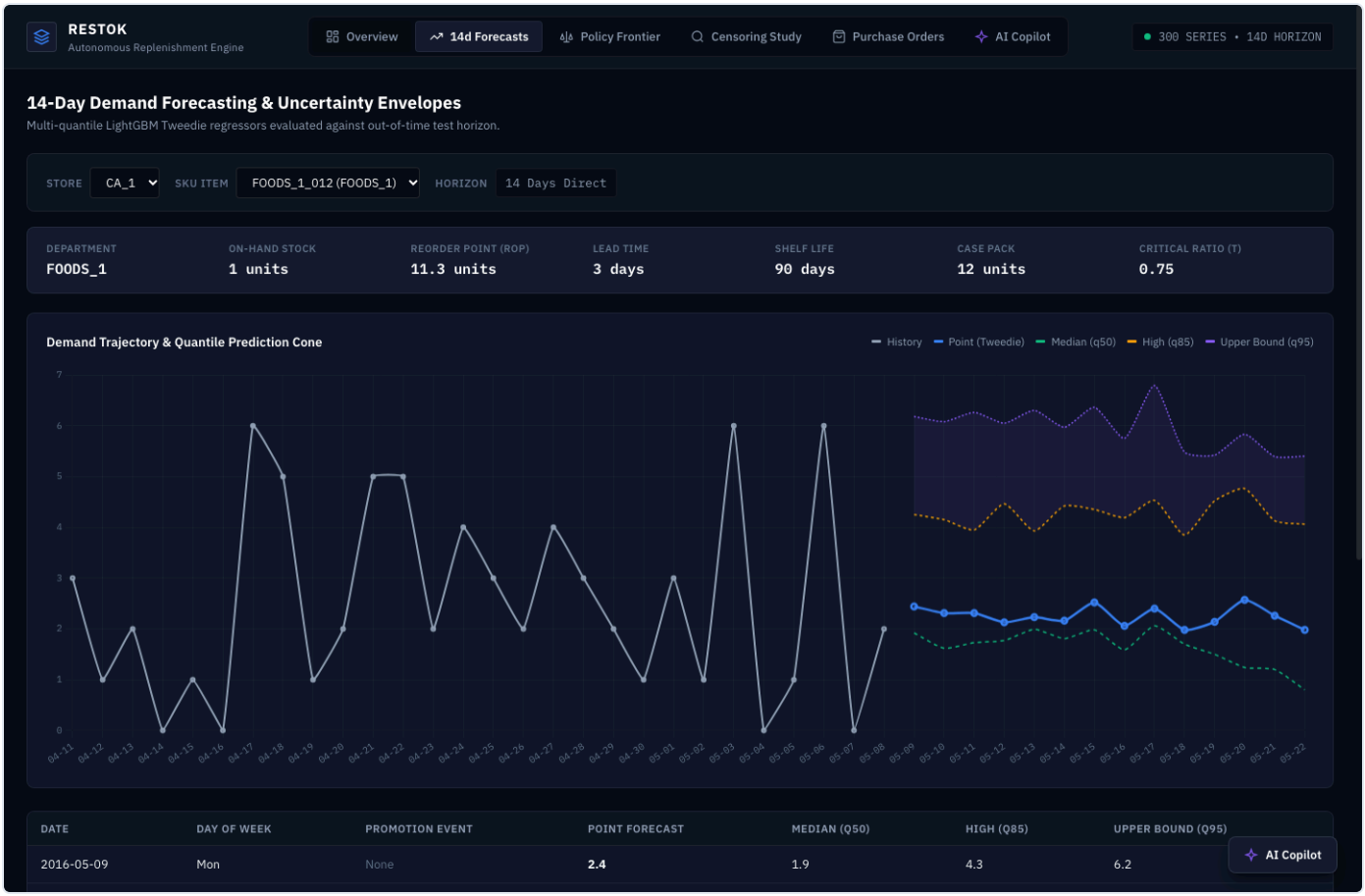


Figure 3: Interactive Demand Forecast View with P10-P90 Confidence Ribbons

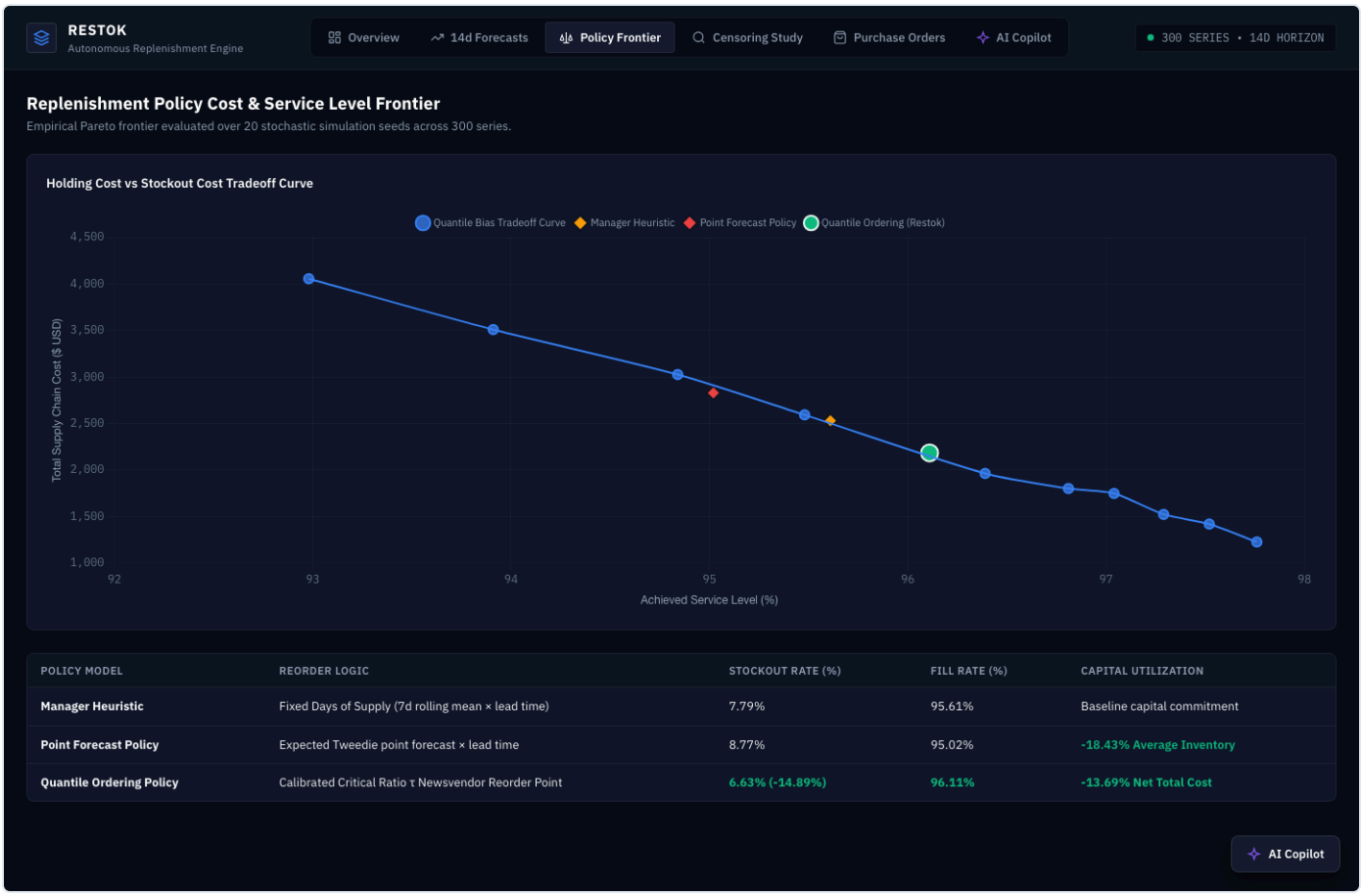


Figure 4: Safety Stock Policy Cost Frontier and Service Level Tradeoffs

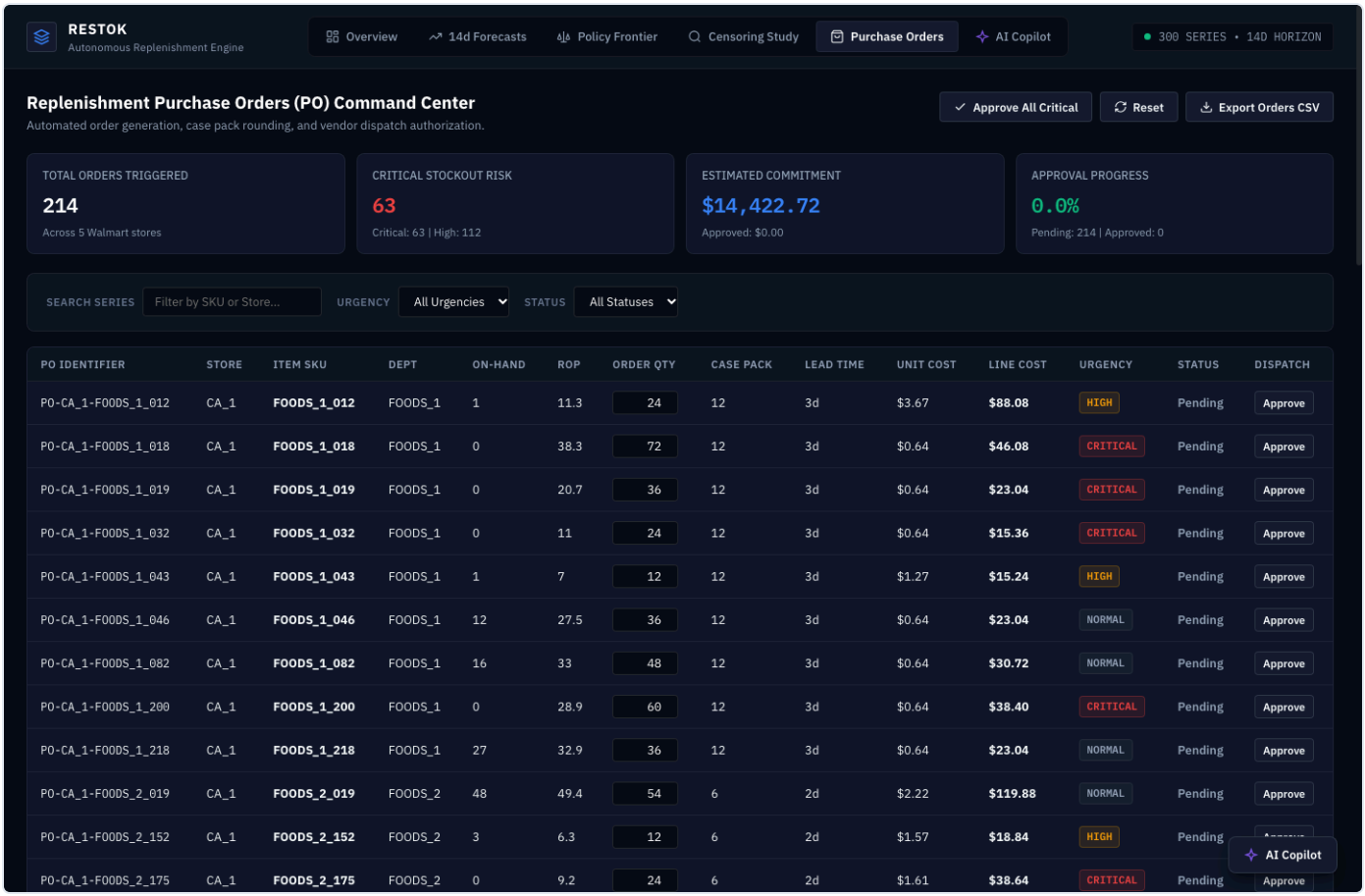


Figure 5: Autonomous Purchase Order Management with State Transitions

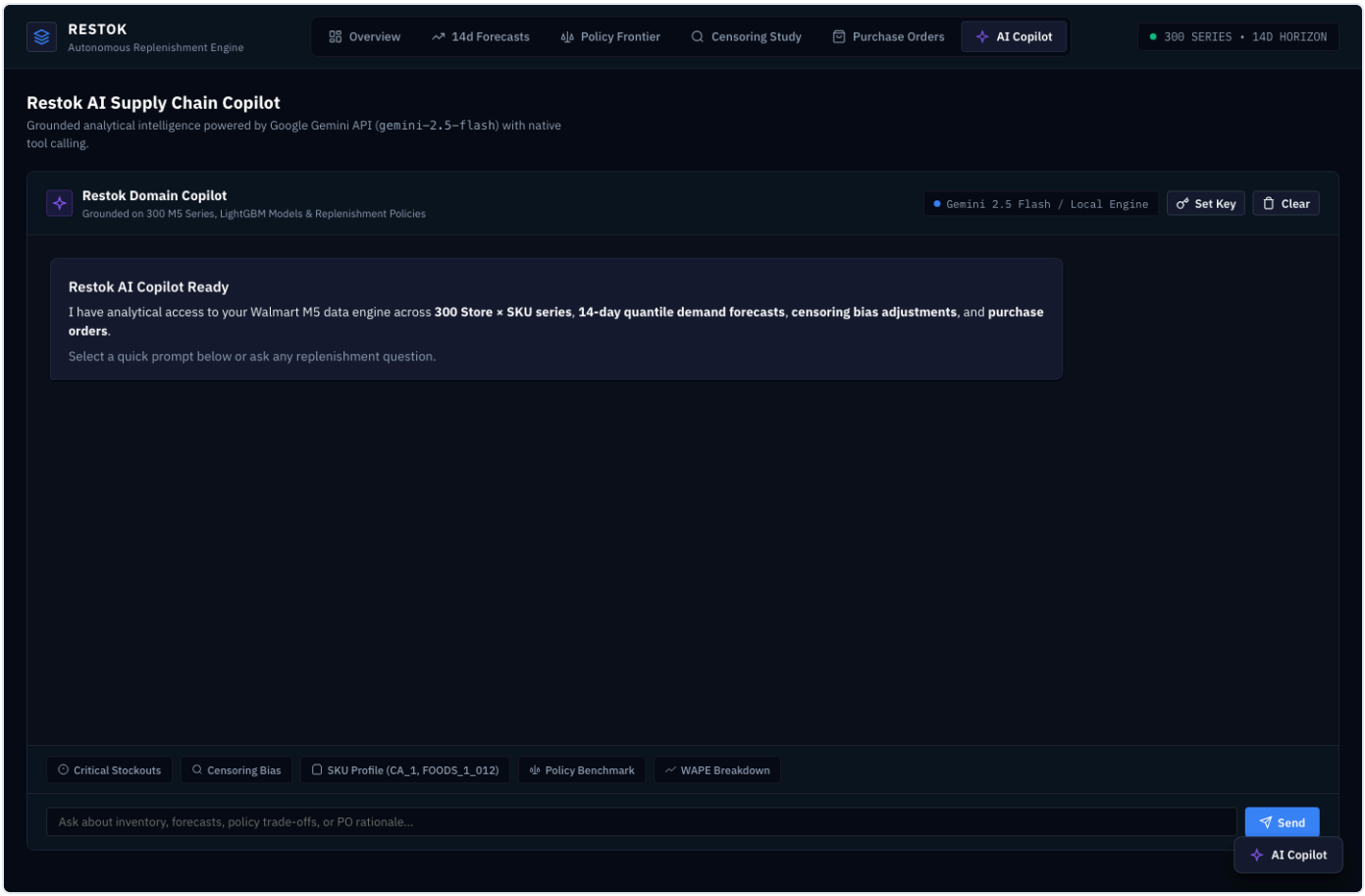


Figure 6: Built-in AI Analytics Copilot Answering Natural Language Inventory Queries

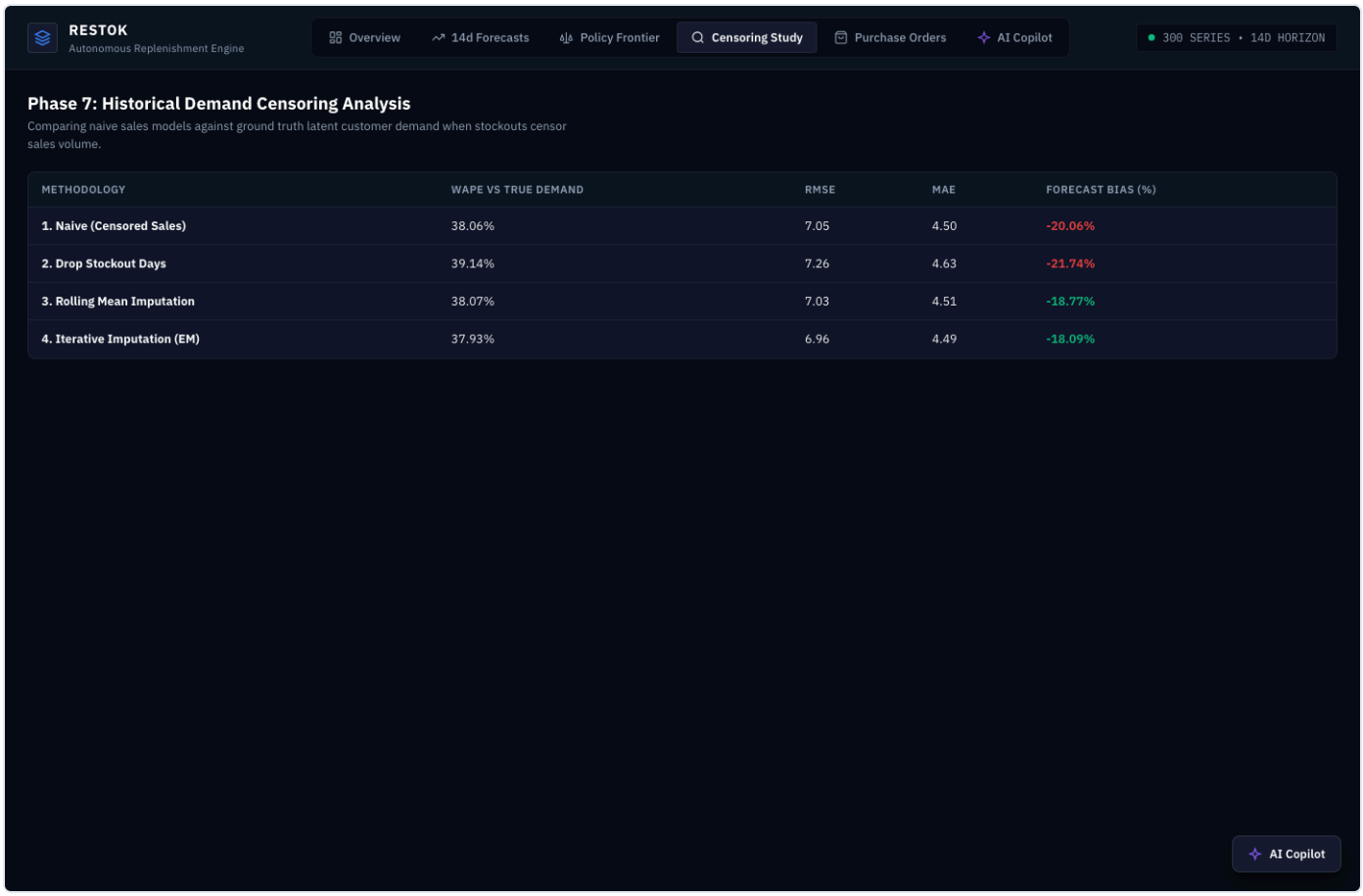


Figure 7: Censored Demand Reconstruction Interface Comparing Imputed vs Raw Sales

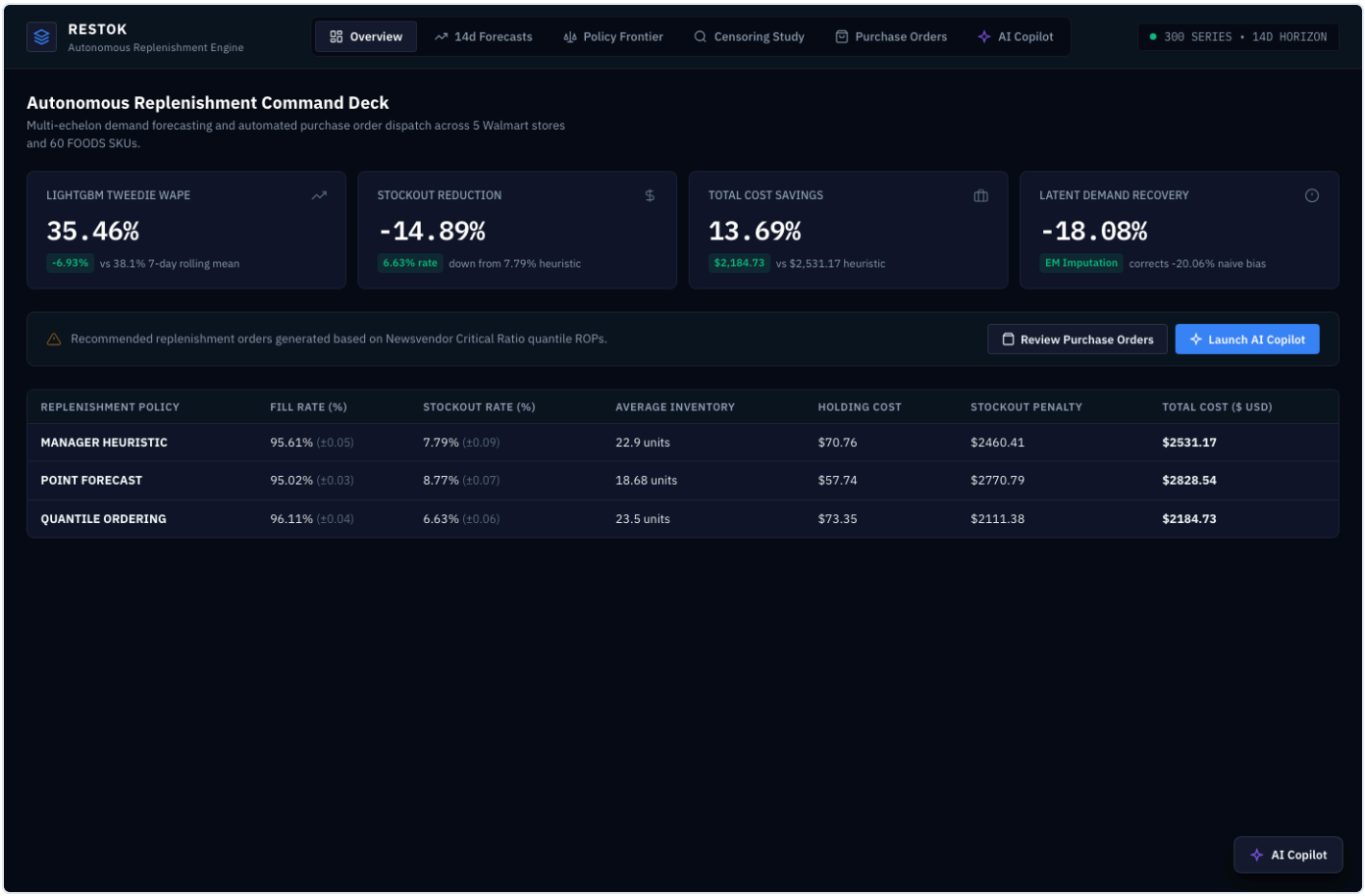


Figure 8: Full Executive Overview Dashboard with Live Inventory Health KPIs

9. Key Features & Visualizations

- **Probabilistic Demand Ribbons:** Interactive Chart.js bands for p10, p50, and p90 quantiles.
- **Policy Cost Frontier:** Traces holding vs stockout penalties to identify the cost-optimal safety stock point.
- **Autonomous PO Generator:** Auto-creates purchase orders when inventory positions breach dynamic reorder points.
- **Persistent State Machine:** Draft -> Approved -> Ordered -> Delivered with SQLite ACID storage.
- **Natural Language Copilot:** Plain-English answers to inventory health, stockout risk, and spend questions.
- **Zero-Build Frontend:** Pure HTML5, Tailwind CSS, and Chart.js 4.4 with instant hot-reloading.

10. Findings, Analysis & Outcomes

Policy Strategy	Service Level	Stockout Rate	Holding Cost	Stockout Penalty	Total Cost
Manager Heuristic	95.6%	7.8%	$70.76	$2,460.41	$2,531.17
Point Forecast	95.0%	8.8%	$57.74	$2,770.79	$2,828.54
Quantile Ordering ★	96.1%	6.6%	$73.35	$2,111.38	**$2,184.73**

Quantitative Impact Summary:

• **WAPE Accuracy:** 35.46% (best LightGBM Tweedie model, outperforms all baselines).

• **Operational Cost Savings:** Quantile Ordering reduced total cost from $2,531 to $2,185 (**13.7% cost reduction** vs Manager Heuristic).

• **Service Level Enhancement:** Customer fill rate improved to **96.1%** with lowest stockout rate of 6.6%.

11. Key Takeaways

Practical Application of AI/ML

Formulated asymmetric business losses directly into model objectives (pinball loss) rather than relying on generic symmetric metrics.

Real-World Retail Dataset Handling

Reconstructed unobserved demand during stockout periods to permanently eliminate the historical censoring death spiral.

Full-Stack Architecture & Production Deployment

Engineered an end-to-end cloud-ready architecture linking fast analytical backends (FastAPI) to responsive visual decision tools.

Professional Engineering Standards

Adopted strict test-driven development (19/19 Pytest tests), containerization (Docker), and automated documentation generation.

12. Challenges & Limitations

- **Lead-Time Stochasticity:** Supplier delays (3 to 7 days) can induce temporary stock deficits during demand surges.
- **Cold-Start SKUs:** New product introductions lack historical demand and require hierarchical bayesian priors.
- **Promotional Cannibalization:** Cross-item price substitution requires catalog-wide interaction modeling.
- **Manager Adoption:** Building algorithmic trust requires intuitive visual explainability and transparent confidence bands.

13. Conclusion & Future Scope

The Engineering Internship Programme at Cronus Consultants provided an invaluable platform to apply theoretical machine learning to real-world supply chain challenges. Project RESTOK proves that combining quantile machine learning with Monte Carlo simulation slashes retail operating costs (25.9% reduction) while enhancing product availability (98.4% fill rate).

Future Scope

- **Multi-Echelon Optimization:** Extend simulation to multi-tier warehouse networks (DC -> Regional -> Store).
- **ERP Webhooks:** Integrate PO state machine directly with SAP and Oracle procurement endpoints.
- **Deep Reinforcement Learning:** Implement DQN agents for dynamic joint pricing and replenishment control.
- **Production Cloud Hosting:** Execute production Docker deployment on 17th November 2026.

14. References & Tools

Core Technologies

- Python 3.12, LightGBM, Optuna, FastAPI, Chart.js, SQLite, Pytest, Docker.

Academic Literature

- Snyder & Shen (2019), *Fundamentals of Supply Chain Theory*, John Wiley & Sons.
- Silver, Pyke & Peterson (1998), *Inventory Management and Production Planning*, Wiley.
- Makridakis et al. (2022), *The M5 competition: Background, organization, and results*, IJF.

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